

Practical 4

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In [1]: import tarfile
import pandas as pd
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.model_selection import train_test_split

# Define a filter function to allow only regular files to be extracted
def tar_safe_filter(tarinfo, path=None):
    if tarinfo.isreg(): # Check if it's a regular file
        return tarinfo # Include this file in extraction
    else:
        return None # Skip non-regular files
# Open and extract the .tgz file using the filter argument
with tarfile.open("cal_housing.tgz", "r:gz") as tar:
    tar.extractall(path=".", filter=tar_safe_filter)

# Load the dataset from the extracted file
california_data = pd.read_csv("cal_housing.data", header=None, delimiter=",")

# Assign column names
california_data.columns = [
    "MedInc", "HouseAge", "AveRooms", "AveBedrms", "Population",
    "AveOccup", "Latitude", "Longitude", "MedHouseVal"
]

# Separate features and target variable
X_reg = california_data.drop('MedHouseVal', axis=1)
y_reg = california_data['MedHouseVal']

# Split the data
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X_reg_train, X_reg_test, y_reg_train, y_reg_test = train_test_split(X_reg, y_reg, test_size=0.3, random_state=42)

# Transform features to polynomial features
poly = PolynomialFeatures(degree=2)
X_poly_train = poly.fit_transform(X_reg_train)
X_poly_test = poly.transform(X_reg_test)

# Train the model
poly_reg = LinearRegression()
poly_reg.fit(X_poly_train, y_reg_train)

# Predict and evaluate
y_poly_pred = poly_reg.predict(X_poly_test)
mse_poly = mean_squared_error(y_reg_test, y_poly_pred)
r2_poly = r2_score(y_reg_test, y_poly_pred)
print(f"Polynomial Regression - MSE: {mse_poly}, R2 Score: {r2_poly}")
```

Polynomial Regression - MSE: 3983828403.4002914, R2 Score: 0.6964794919649606